

Sangjin Choi

Postdoctoral Researcher

CASYS Lab, Information & Electronics Research Institute, KAIST
sjchoi@casys.kaist.ac.kr

RESEARCH STATEMENT

I build cost-efficient AI infrastructure by designing systems software around hardware capabilities and AI workload behavior.

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) Ph.D. in School of Computing Advisor: Youngjin Kwon, CASYS Lab	Daejeon, South Korea 2021-09 – 2026-08
Korea Advanced Institute of Science and Technology (KAIST) Master of Science in School of Computing Advisor: Youngjin Kwon, CASYS Lab	Daejeon, South Korea 2019-09 – 2021-08
Korea Advanced Institute of Science and Technology (KAIST) Bachelor of Science in School of Computing	Daejeon, South Korea 2015-03 – 2019-08

WORK EXPERIENCE

Korea Advanced Institute of Science and Technology (KAIST) Postdoctoral Researcher, Information & Electronics Research Institute	Daejeon, South Korea 2026-09 – Present
Microsoft Research Asia Research Intern, Vancouver Lab	Vancouver, Canada 2026-03 – 2026-05
Microsoft Research Asia Research Intern, Systems and Networking Research Group	Beijing, China 2025-07 – 2026-01

PUBLICATIONS

- [1] [ACM ATC 2026] **Sangjin Choi**, Sukmin Cho, Yifan Xiong, Ziyue Yang, Youngjin Kwon, and Peng Cheng. “ELDR: Expert-Locality-Aware Decode Routing for PD-Disaggregated MoE Serving.”
- [2] [MLSys 2026] Hyunjae Lee, **Sangjin Choi**, Seungjae Lim, and Youngjin Kwon. “BEAM: Joint Resource–Power Optimization for Energy-Efficient LLM Inference under SLO Constraints.”
- [3] [ACM EuroSys 2026] Changjun Lee, **Sangjin Choi**, and Youngjin Kwon. “MTTM: Dynamic Fast Memory Partitioning with Bandwidth Optimization for Multi-tenant Cloud.”
- [4] [Findings of NAACL 2025] Sukmin Cho, **Sangjin Choi**, Taeho Hwang, Jeongyeon Seo, Soyeong Jeong, Huije Lee, Hoyun Song, Jong C. Park, and Youngjin Kwon. “Lossless Acceleration of Large Language Models with Hierarchical Drafting based on Temporal Locality in Speculative Decoding.”
- [5] [IEEE/ACM ICCAD 2023] Jaehoon Heo, Yongwon Shin, **Sangjin Choi**, Sungwoong Yune, Jung-Hoon Kim, Hyojin Sung, Youngjin Kwon, and Joo-Young Kim. “PRIMO: A Full-Stack Processing-in-DRAM Emulation Framework for Machine Learning Workloads.”
- [6] [USENIX ATC 2023] **Sangjin Choi**, Inhoe Koo, Jeongseob Ahn, Myeongjae Jeon, and Youngjin Kwon. “EnvPipe: Performance-preserving DNN Training Framework for Saving Energy.”
- [7] [USENIX ATC 2022] **Sangjin Choi**^{*}, Taeksoo Kim^{*}, Jinwoo Jeong, Rachata Ausavarungnirun, Myeongjae Jeon, Youngjin Kwon, and Jeongseob Ahn. “Memory Harvesting in Multi-GPU Systems with Hierarchical Unified Virtual Memory.” (*Equal contribution.)